ZCAS University

CCS2301 – OPERATING SYSTEMS CONCEPTS

MID-SEMESTER TEST

Thursday, 23rd April 2026

10:30 Hrs – 12:00 Hrs

TIME ALLOWED: WRITING – 1.5 HOURS

READING TIME – 5 MINUTES

INSTRUCTIONS:

1. All the twenty multiple-choice questions in **Section One** (Question One) are **compulsory** and must be attempted. ONLY a letter A, B, C, or D should be indicated as an answer for each multiple-choice question. NO explanations are required. Indicating 2 or more letters on a question will result in a zero mark for that question.
2. **Section Two** has two questions (Question Two and Question Three), **ONLY ONE** of the two questions should be attempted.
3. Maximum score for each Section is 20 marks: Giving a total of 40 marks for the whole test.
4. Numbering of answers in your answer booklet should be the same as that in the question paper. Answers that are not numbered will NOT be marked.
5. Use of scientific calculators during the test is NOT allowed.
6. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION ONE – Answer ALL the 20 Multiple Choice Questions
(Each question carries one (1) mark, giving a total of 20 Marks)

1. What is an operating system?
   A. Application software
   B. A type of computer hardware
   C. Software that manages computer hardware and software resources
   D. A type of memory

2. Which function is not performed by the operating system?
   A. Memory management
   B. Process Management
   C. Virus protection Management
   D. File management

3. What does OS virtualization primarily accomplish?
   A. Enhances the graphic interface
   B. Allows multiple OS to run on the same hardware
   C. Speeds up the processor
   D. Expands RAM size virtually

4. Which component is directly interfaced by the operating system to perform its tasks?
   A. The user
   B. The hardware
   C. The software application
   D. External networks

5. What is the kernel of an operating system?
   A. The interface between the user and the hardware
   B. The central component of the operating system that manages operations
   C. Only responsible for data storage on the disk
   D. The program that controls networking tasks

6. Device drivers are used to:
   A. Increase the speed of devices
   B. Help the operating system communicate with hardware devices
   C. Interface with users
   D. Manage files and directories

7. What is a process in the context of operating systems?
   A. An algorithm
   B. Steps undertaken to achieve a target
   C. A type of operating system
   D. An instance of a program in execution

8. A thread is:
   A. A lightweight process
   B. Another term for a program
   C. A type of process that cannot execute concurrently
   D. A component of an operating system

9. The main purpose of a Process Control Block (PCB) is to:
   A. Store the process executable code
   B. Keep track of process states
   C. Interface with the user directly
   D. Enhance the processing speed

10. Multiprocessing in an operating system allows:
   A. Only one process to execute at one time
   B. Similar to multitasking
   C. Execution of processes in a sequential manner
   D. Multiple processes to execute concurrently

11. Parallel processing is used to:
   A. Reduce the number of CPUs used
   B. Prevent multitasking
   C. Increase the computation power
   D. Decrease system performance

12. Load balancing is important in a multiprocessor system because it:
   A. Ensures all processors carry out the same operations
   B. Distributes tasks evenly across all processors
   C. Prevents any processor from performing tasks
   D. Reduces the need for processors

13. What does preemptive scheduling allow?
   A. A process to run without interruption
   B. The operating system to interrupt a currently running process and to start or resume another
   C. Only one process to run at all times
   D. Transferring a process to virtual memory

14. How does a preemptive scheduling algorithm differ from a non-preemptive one?
   A. It allows a task to be suspended and resumed
   B. It does not allow task prioritization
   C. It only uses a single queue
   D. It cannot handle multiple processes

15. An operating system's scheduler dispatches tasks by switching tasks in and out of the CPU. What is this known as?
   A. Context switching
   B. Task synchronization
   C. Swapping
   D. Task execution

16. What is the key advantage of multithreading?
   A. Reduces memory usage
   B. Improves CPU utilization
   C. Eliminates the need for memory
   D. Simplifies programming

17. Swapping, in the context of an operating system, is:
    A. Exchanging data between RAM and the hard disk to free up memory
    B. Switching users in a multi-user environment
    C. Exchanging process priorities
    D. Changing a program into a process

18. Which type of operating system interface uses typed commands?
    A. Graphical User Interface
    B. Command Line Interface
    C. Touch Interface
    D. Operating System Kernel

19. What is a multi-user operating system?
    A. An OS that allows only one user at a time
    B. An OS that supports one program at a time
    C. An OS that supports multiple users simultaneously
    D. An OS that does not interact with the hardware

20. What is the role of the long-term scheduler?
    A. Select processes from ready queue
    B. Select processes to be loaded into memory
    C. Handle I/O scheduling
    D. Manage CPU registers

SECTION TWO – Answer ONE of the 2 Questions below
(Each question carries twenty (20) marks)

QUESTION TWO
NOTE: NO essay type of answering. Just go to the point and answer each question separately.

i) State the difference between the following terms: **(10 Marks)**
    a) A Volatile Device and a Primary Memory Device
    b) A Process and a Thread
    c) Multiprocessing and Multithreading
    d) Preemptive Scheduling and Non-Preemptive Scheduling
    e) Swapping and Context Switching
ii) Explain the function of each of the following components of an Operating System:
    f) Short-Term Scheduler **(6 Marks)**
    g) Medium-Term Scheduler
    h) Long-Term Scheduler
iii) State the four (4) components / segments of a process. **(4 Marks)**

QUESTION THREE
The diagram below summarizes various states that a process may change into during its life cycle. Using the diagram or otherwise, you are required to answer the ten (10) questions under Part One and Part Two below.
NOTE: NO essay type of answering. NO justification or explanations. Just state.

Question 3 – Part One

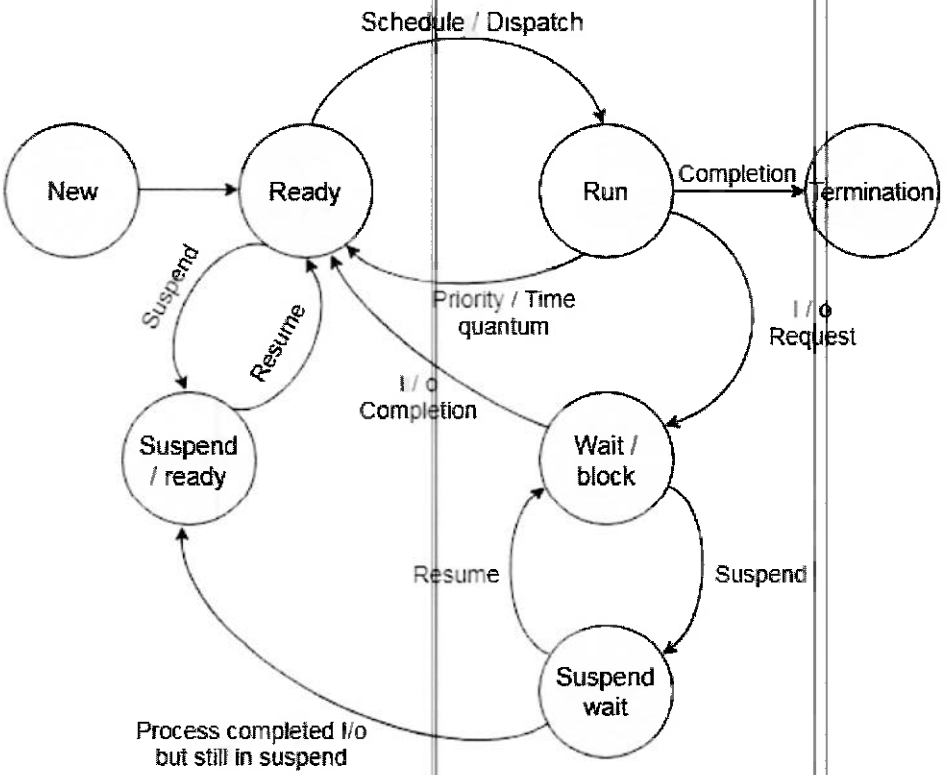
State any ONE condition or situation that may be the reason as to why a particular process would be made to change from one state to another as indicated below. **(10 Marks)**

i) From the Running State to the Ready State
ii) From the Running State to the Waiting State
iii) From the Ready State to the Suspend-Ready State
iv) From the Suspend-Wait State to the Suspend-Ready State
v) From the Suspend-Wait State to Waiting State

Question 3 – Part Two

State a component / part of a computer system where you would expect to find processes that are in each of the following state: **(10 Marks)**

vi) Processes in a Running State.
vii) Processes in a Suspend-Ready State.
viii) Processes in a Ready State.
ix) Processes in a Waiting State.
x) Processes in a New State



END OF TEST